

Linear filtering: Separable filtering, Band -pass and steerable filters

Why Do We Need Filters in Image Processing?

Images contain:

- Useful information → edges, textures, shapes, objects
- Undesired components → noise, blur, illumination variations

Filters help us:

- Enhance important features
- Suppress irrelevant information
- Prepare images for higher-level tasks (segmentation, recognition, tracking)

Linear filters are preferred because they are:

- Mathematically tractable
- Computationally efficient
- Easy to analyze in both spatial and frequency domains

1. Separable Filters

What Is a Separable Filter?

A filter is separable if a 2D convolution can be broken into two 1D convolutions:

Instead of filtering the image in 2D directly, we:

1. Filter along rows
2. Filter along columns

Why Are Separable Filters Used?

1. Computational Efficiency

- 2D convolution with kernel → operations per pixel
- Separable filter → operations per pixel
- This is **critical for real-time systems**

2. Memory and Hardware Efficiency

- Fewer multiplications and additions
- Ideal for embedded systems, GPUs, and mobile vision applications

3. Mathematical Simplicity

- Many important filters (Gaussian, Sobel) are naturally separable
- Enables efficient multi-scale processing

Common Examples

- Gaussian smoothing filter
- Sobel edge detector
- Prewitt filter

Advantages

- Very fast computation
- Reduced computational complexity
- Easy to implement
- Suitable for large kernel sizes

Limitations

- Not all filters are separable
- Limited ability to model complex 2D patterns
- Orientation selectivity is weak unless combined with other filters

Applications

- Image smoothing and denoising
- Gradient and edge computation
- Preprocessing in CNNs
- Medical image smoothing
- Video processing pipelines

2. Band-Pass Filters

What Is a Band-Pass Filter?

A band-pass filter allows only a specific range of frequencies to pass while blocking:

- Low frequencies → illumination, background
- High frequencies → noise

It isolates mid-frequency information, which often corresponds to:

- Edges
- Textures
- Repetitive patterns

Why Are Band-Pass Filters Used?

1. Feature Enhancement

- Most meaningful visual structures lie in specific frequency bands
- Band-pass filtering enhances these structures

2. Noise and Illumination Suppression

- Low-frequency lighting variations are removed
- High-frequency noise is suppressed

3. Scale Selectivity

- Objects appear at different scales
- Band-pass filters can be tuned to specific sizes

Common Band-Pass Filters

Difference of Gaussians (DoG)

- Subtracts two blurred images
- Approximates Laplacian of Gaussian
- Efficient and widely used

Laplacian of Gaussian (LoG)

- Detects zero crossings corresponding to edges
- Strong theoretical foundation

Gabor Filters

- Jointly selective to frequency and orientation
- Inspired by human visual cortex

Advantages

- Excellent for texture and edge analysis
- Multi-scale analysis possible
- Good balance between noise suppression and detail preservation

Limitations

- Parameter selection (scale, frequency) is critical
- Sensitive to scale mismatch
- Computationally more expensive than simple smoothing

Applications

- Edge detection
- Texture classification
- Face and fingerprint recognition
- Medical imaging (tumor boundary detection)
- Remote sensing and satellite imagery
- Feature extraction in machine learning

3. Steerable Filters

What Is a Steerable Filter?

A steerable filter can be oriented to any angle by combining a small set of basis filters:

No physical rotation of the filter is required.

Why Are Steerable Filters Used?

1. Orientation Selectivity

- Real-world images contain edges at arbitrary orientations
- Fixed-direction filters miss many features

2. Computational Efficiency

- Instead of rotating filters many times, responses are synthesized mathematically
- Reduces redundancy and computation

3. Robust Feature Representation

- Provides continuous orientation information
- Improves rotation invariance

Advantages

- Precise orientation tuning
- No interpolation artifacts from rotation
- Fewer filters needed
- Ideal for multi-orientation analysis

Limitations

- Requires pre-defined basis filters
- More complex mathematical formulation
- Higher memory overhead for storing basis responses

Applications

- Edge and contour detection
- Corner and junction detection
- Texture segmentation
- Object and shape recognition
- Medical image analysis (vessel detection)
- Autonomous driving (lane and boundary detection)

Comparison:

Aspect	Separable	Band-Pass	Steerable Filters
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	Filters	Filters	
Purpose	Efficient filtering	Frequency selection	Orientation selectivity
Key strength	Speed	Feature isolation	Directional analysis
Complexity	Low	Medium	Medium–High
Selectivity	Spatial	Frequency	Orientation
Inspired by	Math optimization	Signal processing	Human vision

Example:

1. Separable Filter – Numerical Example

Given Image (3×3)

Given Separable Filter

Horizontal filter:

Vertical filter:

Answer:

Step 1: Horizontal Filtering (Row-wise)

Apply to each row.

Row 1:

Row 2:

Row 3:

Intermediate Image

Step 2: Vertical Filtering (Column-wise)

Column 1 Filtering

Column 2 Filtering

Column 3 Filtering

Output After Vertical Filtering

Vertical Filtering for Edge Detection

Vertical Sobel Filter

Apply to Column

2. Band-Pass Filter – Numerical Example (Difference of Gaussians)

Given Signal (1D for simplicity)

Input Signal

Small Gaussian Kernel

Step-1 Kernel (Small Gaussian / Average)

Kernel length = 3

Kernel covers	Calculation	Output
[2, 4, 6]	$(2+4+6)/3$	4
[4, 6, 8]	$(4+6+8)/3$	6
[6, 8, 10]	$(6+8+10)/3$	8

Hence Step-1 Output

Step-2 Kernel (Large Gaussian / Average)

Kernel length = 5

Sliding the Kernel

Only one position exists where the kernel fully fits:

Kernel covers	Calculation	Output
[2, 4, 6, 8, 10]	$(2+4+6+8+10)/5$	6

Hence Step-2 Output

Step 3: Difference of Gaussians

At center position:

3. Band-Pass Filter – Image Example (Edge Detection)

Image Patch

$$I = \begin{bmatrix} 5 & 5 & 5 \\ 5 & 10 & 5 \\ 5 & 5 & 5 \end{bmatrix}$$

Laplacian Kernel

$$L = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Convolution at Center Pixel

$$\begin{aligned} (5 \times 0) + (5 \times -1) + (5 \times 0) + (5 \times -1) + (10 \times 4) + (5 \times -1) + (5 \times 0) + (5 \times -1) + (5 \times 0) \\ = -5 - 5 + 40 - 5 - 5 = \boxed{20} \end{aligned}$$

4. Steerable Filter – Numerical Example

Given Gradient Responses

Horizontal gradient:

Vertical gradient:

Find Response at

Steerable Formula

Calculation

Interpretation

- Edge strength is strongest at 45°

- **No need to rotate filter**